The Earthquake Swarm at the Onset of the 7 June 2023 Kīlauea Eruption: Mechanisms of the M 3.3 Onset Earthquake and Its Neighbors

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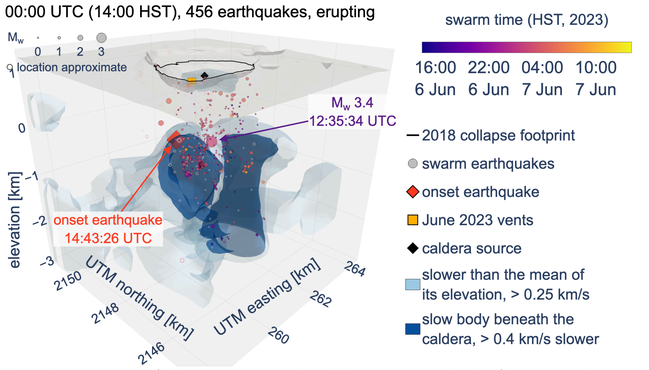
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QR to the swarm movie

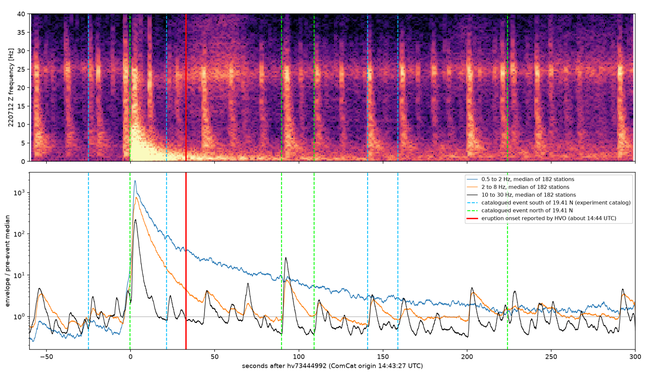
THE SWARM AND THE ERUPTION

- Kīlauea erupted in Halemaʻumaʻu on 7 June 2023 at about 14:44 UTC after a 14 h earthquake swarm under the caldera
- The last earthquake before the onset, **M 3.3 at 14:43:27 UTC**, is about 100 times larger than any other event of the swarm
- 182 stations within 10 km recorded it: 89 GTI and 9 PASSCAL nodes of our array, 70 HVO SmartSolo nodes, 14 HV broadband stations
- We ask what the earthquakes of the swarm did, in which direction the rock moved, and whether the onset earthquake differs from the rest



The relocated swarm (456 earthquakes, 00:34 UTC to the next midnight) in the ambient-noise V_S volume, dots sized by M_w and colored by time; the June 2023 vents in orange; one frame of the movie behind the QR code

THE ONSET IN THE DATA



Six minutes around the M 3.3 on node 220712, 2 km south: spectrogram above, network-median envelopes in three bands below; dashed lines are catalogued events (blue south of 19.41 N, green north), the red line the onset reported by HVO

- Small events every 10 to 30 s run straight through the M 3.3 and the onset
- Within 2 to 3 min a low-frequency tremor rises from under the coda and keeps growing while the high-frequency events thin out
- No low-frequency precursor on any station; the burst 4 s before the M 3.3 is a small, shallow earthquake 2 km south

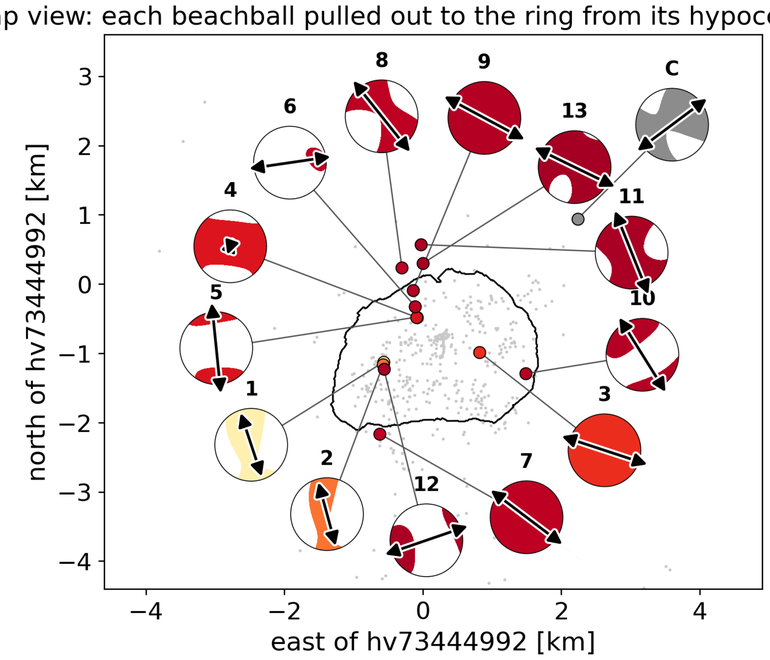
WHAT WE DID

- **Velocity model:** the T-REX V_P of O'Connell, Girard et al. (Part I, in prep.) with the ambient-noise V_S of Girard, O'Connell, Shragge et al. (Part II); two V_S volumes, group-velocity and phase-only, bracket the uncertainty
- **Green's functions:** SPECFEM3D on the 25 m model, 400 m mesh with topography and the 2023 pit, one run per source element and position
- **Inversion:** MTUQ, full moment tensor and double couple compared by BIC; P and S windows at 0.5 to 0.8 Hz, 14 earthquakes (M 2.0 to 3.7), 3 by 3 position grids at 300 m for the two largest
- **Reading a mechanism:** the filled part of each ball is where the first motion pushed outward; the double arrow is the direction of greatest extension; shear/mixed also covers events whose data cannot tell

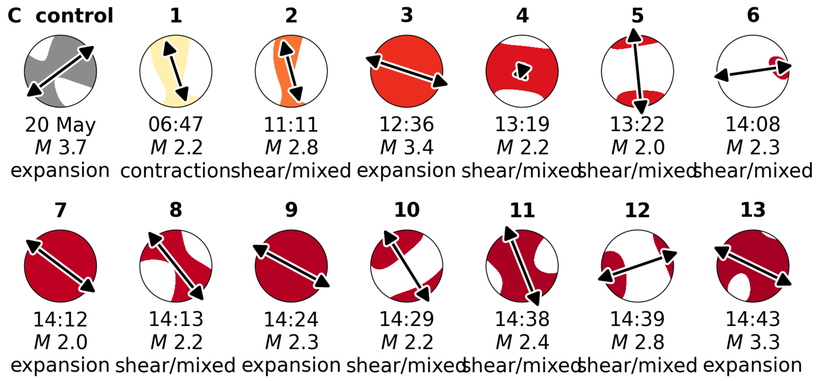


MECHANISMS OF THE SWARM, PULLED APART

map view: each beachball pulled out to the ring from its hypocenter (dot)



the same mechanisms in time order (UTC, 7 June 2023), model A

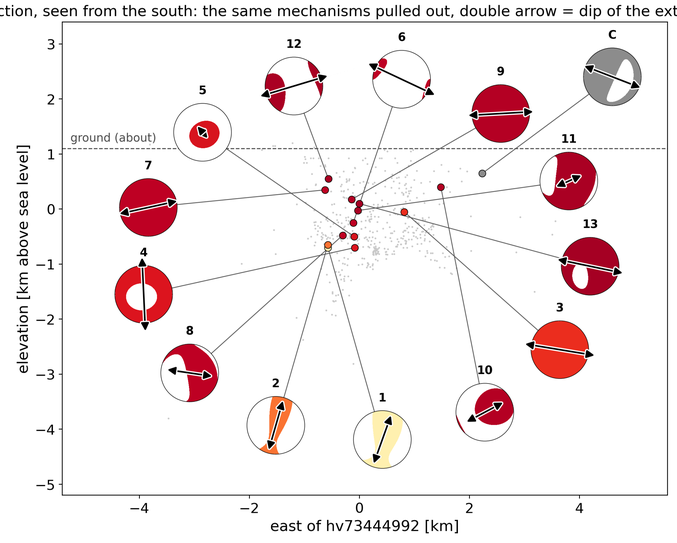


filled: compression, in the color of the time; double arrow: direction of extension (tension axis) in map view; gray dots: the relocated swarm; black line: 2018 collapse

Left: every mechanism pulled out of the cluster onto a ring, a gray line to its hypocenter (dot); filled is compression, in the color of the time; the double arrow is the direction of greatest extension seen from above; C is a control earthquake of 20 May with no eruption. Right: the same mechanisms in time order with their plain reading. The three largest events (3, 9, 13) extend WNW to ESE; the onset earthquake (13) and the control read the same way

THE SAME MECHANISMS IN DEPTH

east-west depth section, seen from the south: the same mechanisms pulled out, double arrow = dip of the extension direction (model A)

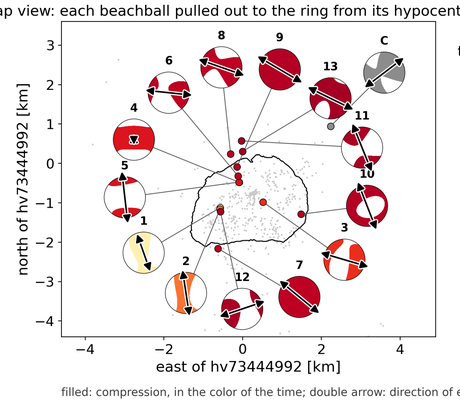


numbers as in the map figure (time order; C the 20 May control); filled: compression toward the viewer, in the color of the time; gray dots: the relocated swarm

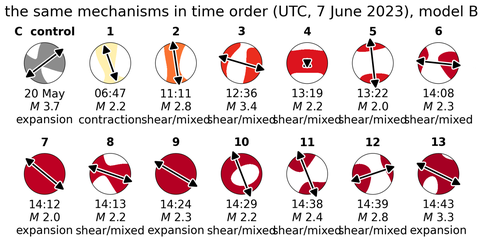
The same mechanisms in an east-west depth section seen from the south, numbered as on the map; here the double arrow shows how steeply the extension direction dips. The largest events extend nearly horizontally; the small events scatter

THE SAME THROUGH A SECOND VELOCITY MODEL

map view: each beachball pulled out to the ring from its hypocenter (dot)



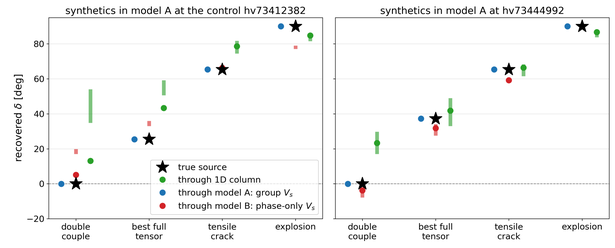
the same mechanisms in time order (UTC, 7 June 2023), model B



filled: compression, in the color of the time; double arrow: direction of extension (tension axis) in map view; gray dots: the relocated swarm; black line: 2018 collapse

The same through the second V_S volume (B, phase-only): the directions of the three largest events and the control hold; the M 3.4 of 12:36 (3) becomes shear/mixed at B's best position

CAN THE ARRAY TELL EXPANSION FROM SHEAR?



Known sources (stars) fed through the same chain: blue through the right model, red through the second V_S, green through a 1D model

- **What it tests:** whether the array and the 3D model can tell expansion from shear at all
- **What it shows:** with the right 3D structure the recovered source lands on the star; the second V_S moves it by at most 6° a 1D model turns a pure shear source into 13 to 23° of apparent expansion
- **Why it matters:** the control earthquake reads as expansion too, by about that amount, so the expansion sign of the onset earthquake is not claimed; the directions of extension, which the 1D test leaves intact, are

CONCLUSIONS AND NEXT

- A dense array at the vent with a 3D model fits the P and S waves of M 2 to 3.7 earthquakes and gives a mechanism for each
- The full moment tensor beats a double couple for both large onset events and for the control; the large events extend WNW to ESE, near horizontally
- The amount of expansion is within what an imperfect model produces, so its sign is left open
- **Next:** the whole swarm once the catalog is relocated in the newest V_P; the directions of extension as the input to a map of the magma path

REFERENCES AND ACKNOWLEDGMENTS

O'Connell, D. R. H., Girard, A. J., et al. (in prep.) Velocity modeling at Kīlauea 2023 Part I: V_P from active-source tri-axial vibroseis

Girard, A. J., O'Connell, D. R. H., Shragge, J., et al. (in prep.) Velocity modeling at Kīlauea 2023 Part II: Low-frequency V_S from tomography, eikonal tomography, HVSR and earthquake relocation

Tape & Tape (2012) GJI 190; Thurin et al. (2025) MTUQ; Komatitsch & Tromp (2002) GJI 149; Mulliken et al. (2024) HVO chronology

Nodes: GTI, PASSCAL, HVO; HV data via EarthScope; computing on Wendian (Mines) and gpu2